

ACHINTYA RAI

925-202-1555 | achintyaraism@gmail.com | linkedin.com/in/achintya-rai | github.com/Azurinine

EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe
Software Engineering Intern

San Jose, CA
June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks with Python to debug the production data source, exposing **86,387** duplicate records and a **47%** timestamp null rate.
- Diagnosed a **26-hour** processing outage across **1,272** monitoring segments via volume analysis.
- Deployed an MLOps pipeline for weekly retraining and hourly inference across **30+** countries.

VALT Health
Founding Software Engineer

Remote
Aug. 2025 – Feb. 2026

- Engineered a Python/Flask backend processing **100k+** datapoints from fitness devices, designing schemas to unify formats across **20+** distinct data sources.
- Optimized async batch processing in GCP for a **35%** speedup and **60%** fewer database writes.
- Authored technical specifications and documentation standards for engineer onboarding.
- Automated environment setup and data seeding via Bash scripts, cutting testing time **80%**.

Decklar (formerly Roambee)
Software Engineering Intern

Sunnyvale, CA
June 2024 – Aug. 2024

- Developed real-time shipment modules deployed for **Amazon, Coca-Cola, and Samsung**, optimizing Python/NumPy pipelines for a **15%** capacity boost on **1M+** weekly events.
- Implemented advanced data structures for analytics modules, delivering **20%** faster reporting.
- Accelerated workflow efficiency **15%** through Panels-based UI refinements and user testing.

RESEARCH

Graph Theory & Neural Architecture
Undergraduate Research Fellow (NSF REU)

San Diego, CA
March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), building extraction utilities that reduced **500GB** checkpoints to **13GB** localized manifolds.
- Optimized placement scripts and preprocessing for a **30x runtime reduction** on **8GB RAM** systems.
- Applied **Rent's Rule** and Louvain clustering, raising Inter-Cluster density to **13.25%**.

PROJECTS

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker, CI/CD* 

Aug. 2025 – Dec. 2025

- Deployed a full-stack coaching platform on **AWS** using **React, TypeScript, and WebSockets**.
- Architected **20+ Django REST API endpoints** for authentication, sessions, and transactions.
- Established a **CI/CD pipeline** with automated testing and error logging for **99.9%** reliability.

MovieRecapinator | *Java, Spring Boot, Maven, JavaScript, IMDb API* 

April 2025

- Built a Java/**Spring Boot** engine processing **700,000+** IMDb entries to rank recap episodes.
- Collaborated with a **4-person** team to ship a full-stack MVP in Java and JavaScript in **24 hours**.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Development: React, Django, Node.js, FastAPI, WebSockets, Flask, REST APIs, JUnit, PyTest, CI/CD

AI & Cloud: PyTorch, TensorFlow, AWS, GCP, Docker, GitHub Actions, Linux, Gemini/OpenAI API, MCP